

Slide 1: Title & Project Overview

DESIGN & IMPLEMENTATION OF A HIGH-FIDELITY PHARMACY INVENTORY & PRESCRIPTION COMPLIANCE DATABASE SYSTEM IN THE KYRGYZ REPUBLIC

COMP2082: Advanced Relational Databases & DBMS Architectures

Presenter: Rizvon Navruzov

Academic Advisor: Nurlan Shaidullaev

Slide 2: The Problem Statement

Core Challenges in Community Retail Pharmacy Operations:

- **Inventory Spoilage & Expiry Leakage:** Dispensing expired stock due to lack of batch-level tracking.
 - **Prescription Evasion:** Selling regulated drugs (antibiotics/psychotropics) without a valid clinical script.
 - **Race Conditions:** Concurrent checkouts double-selling the same physical batch, causing inventory imbalances.
 - **Tax Auditing Lag:** Manual calculations of the **standard 12% VAT** in Kyrgyzstan leading to accounting errors.
 - **Security Violations:** Standard cashiers accessing staff salary logs or personal identification keys.
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Slide 3: Entity-Relationship Model (12 Normalized Tables)

Architectural Layout & Cardinalities:

- **Recursive Hierarchies:** Mapped on `categories(parent_category_id)` to allow multi-level drug classifications.

- **Patients & Employees:** Registered with dates of birth and mandatory **14-digit national PINs** (ПИН/ИИН).
 - **M:N Junction Table Resolution:**
 - `prescription_items` resolves: `prescriptions` ↔ `medicines`
 - `sale_items` resolves: `sales` ↔ `batches`
 - **Audit Logging Ledger:** `inventory_adjustments` logs stock counts, breakages, and supplier returns.
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Slide 4: Physical Schema & Type Selections

PostgreSQL 15+ Advanced DDL Implementation:

- **SQL-Standard Identity Columns:** Mapped using `GENERATED ALWAYS AS IDENTITY` to prevent manual sequence bypasses.
 - **Native Custom ENUM Domains:** Implements type constraints without small lookup join tables:
 - `biological_gender ('M', 'F')`
 - `employee_role ('pharmacist', 'manager', 'admin')`
 - `payment_method ('cash', 'card', 'mobile_qr')`
 - `prescription_status ('pending', 'partially_filled', 'filled', 'expired')`
 - `adjustment_type ('breakage', 'spoilage', 'theft', 'reconciliation', 'return_to_supplier')`
 - **Base Currency:** Formatted using `NUMERIC(12, 2)` in **Kyrgyz Som (KGS)**.
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Slide 5: Rigorous Normalization (3NF & BCNF)

Eliminating Database Redundancies and Anomaly Vectors:

- **BCNF Proof:** Proven mathematically for core tables (`medicines`, `batches`). Every non-trivial functional dependency $X \rightarrow Y$ has a determinant X that is a candidate key.
 - **Deliberate Denormalization:** `sales` table stores `total_gross`, `tax_amount`, and `total_net`.
 - *Performance Justification:* Prevents expensive aggregate SUM operations during high-frequency reporting.
 - *Auditing Invariance:* Past receipts remain frozen if historical prices change.
 - *Integrity Shield:* Automatically recalculated on line item changes via triggers.
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Slide 6: PL/pgSQL Automation Engine

Moving Core Business Validation to the Database Layer:

- **tg_deduct_inventory**: Runs on sales checkouts. Locks batch rows via SELECT FOR UPDATE to prevent race conditions, validates non-expiration, verifies quantities, and deducts stock.
 - **tg_validate_prescription**: Checks prescription-only medicines, validates patient/doctor relationships, verifies remaining limits, and auto-updates prescription status.
 - **tg_sale_items_totals**: Automatically extracts the **standard 12% VAT** from gross pricing, updates net amounts, and deducts discounts.
 - **tg_sync_inventory_adjustment**: Captures inventory adjustments (wastage, theft) to keep stock levels reconciled.
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Slide 7: High-Performance Indexing Strategy

Tuning Query Latency to Sub-Millisecond Levels:

- **Live Search B-Tree Index**: Composite indexing of trade and generic names to speed up product searches during checkout:

```
CREATE INDEX idx_medicines_search ON medicines  
(trade_name, generic_name);
```

- **Partial Expiry B-Tree Index**: Optimizes expiry tracking. Excluding depleted batches keeps the index small and lookups fast:

```
CREATE INDEX idx_batches_expiry_tracking ON batches  
(expiry_date, medicine_id) WHERE (current_quantity > 0);
```

- **Transaction Audits**: Mapped B-Tree indexes on sales timestamps and prescription serial numbers.
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Slide 8: BI Analytics Views

Real-Time Dashboard Reporting:

- **v_low_stock_medicines**: Tracks stock levels across active batches and calculates recommended reorder quantities.
- **v_expiring_medicines**: Audits batches expiring within 90 days, calculating tied-up capital and potential losses.

- **v_pharmacy_sales_summary**: Provides monthly financial summaries, calculating net revenue, VAT collected, and order values.
 - **v_patient_prescription_history**: Traces complete patient histories, linking doctors, prescriptions, and remaining refill limits.
 - **v_supplier_medicine_summary**: Analyzes suppliers based on product variety, delivery history, and average acquisition costs.
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Slide 9: ACID Transaction Models & Aborted Rollbacks

Guaranteeing Database State Integrity:

- **Successful Sales Checkout**: Atomic inserts of sales headers and items, with automatic database-level inventory deductions and billing calculations.
 - **Inventory Intake**: Reconciles fresh bulk intake with physical stock and logs adjustments in the audit ledger.
 - **Trigger-Induced Aborted Rollbacks**: Demonstrates **Atomicity**. If a cashier tries to sell more stock than is available, the triggers raise an exception, aborting the transaction and rolling back all changes (including sales headers).
 - **Atomic Clinical Prescribing**: Ensures a prescription and its drug lines are created as a single, indivisible unit to satisfy medical regulations.
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Slide 10: Security Model & Principle of Least Privilege

Group-Level RBAC & Column-Level Protections:

- **Schema Isolation**: Revokes default public access to isolate the schema:

```
REVOKE ALL ON SCHEMA public FROM PUBLIC;
```

- **Privilege Assignments**:
 - **readonly_auditor**: Group role with read-only (SELECT) access across the schema.
 - **cashier**: Can check out sales. Restricted from direct stock adjustments or viewing staff security credentials.
 - **pharmacist**: Can write prescriptions and check out sales. Has column-level access to the employee directory (excluding passwords and PINs).
 - **inventory_manager**: Can manage batches, suppliers, and categories. Restricted from clinical prescriptions and sales checkouts.
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Slide 11: High-Volume Seeding & Queries Demo

Simulating Real-World Pharmacy Operations:

- **3,000+ logically consistent records** generated via PostgreSQL D0 blocks.
 - **100% active triggers** during seeding, validating the trigger logic and automated calculations.
 - **Key BI Queries Supported:**
 - *Dead-Stock Expiration Forecast*: Tracks capital tied up in expiring batches.
 - *Therapeutic Sales Margin Analysis*: Analyzes profitability and COGS by category.
 - *Pharmacist Commission Dashboard*: Calculates average order value (AOV) and cashier sales shares.
 - *Safety Regulatory Auditing*: Audits sales of regulated substances against doctor prescriptions.
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Slide 12: Architectural Reflections & Project Conclusion

Core Design Lessons & Future Roadmap:

- **Database-Level vs. App-Level Checks**: Implementing validation rules in PL/pgSQL triggers ensures consistent compliance across all client apps (web registers, mobile apps, or APIs).
- **Concurrency Locks**: Using `SELECT FOR UPDATE` is crucial for preventing double-selling race conditions in multi-terminal retail checkouts.
- **Future Improvements:**
 - Integrating HL7/FHIR healthcare data standards.
 - Implementing predictive REST APIs for real-time reporting to the Ministry of Health.
 - Adding machine learning restocking models based on seasonal sales trends.